

RITWIK BISWAS

Software Engineer | Backend & AI Systems

+91 83360 83405 | biswas.ritwik2001@gmail.com | LinkedIn | GitHub | Portfolio | Bengaluru, India

EDUCATION

Indian Institute of Technology Kharagpur

2021 - 2026

Dual Degree (B.Tech + M.Tech) in Mechanical Engineering | CGPA: 7.84/10

EXPERIENCE

Taskly Technologies Inc. | Software Engineering Intern

Remote, Canada | Mar 2025 - Jul 2025

- Built core modules for JobSense using Next.js, FastAPI, and PostgreSQL, shipping five AI-assisted workflows for JD analysis, resume scoring, ATS checks, interview preparation, and company research.
- Integrated Groq and Firebase Authentication with JWT-secured user sessions and protected AI endpoints.
- Resolved JavaScript-rendered scraping failures, API payload bottlenecks, environment misconfigurations, and Docker conflicts; containerized services for reliable Vercel releases.

Search-in | Full-Stack Developer Intern

Remote, Bengaluru | Jun 2024 - Jul 2024

- Built a React admin dashboard and Node.js/Express APIs serving 10K+ users, 1K+ orders, and 500+ managed records; automated analytics and operational workflows, increasing engagement by 20%.
- Optimized MongoDB access across high-traffic operations using indexes, Redis caching, and query tuning to reduce database latency and improve concurrent request throughput.
- Redesigned the end-to-end order workflow, reducing processing time by 30%; deployed frontend assets to AWS S3 and cloud-hosted services to improve availability and maintainability.

Pepcorns | Full-Stack Developer Intern

Remote, Pune | Mar 2024 - Apr 2024

- Built a Node.js and SQL referral-rewards system covering three core campaign workflows: referral attribution, reward issuance, and social sharing for end-to-end tracking.
- Developed reusable React components and integrated end-to-end referral journeys with backend APIs and database operations, reducing duplication across user-facing interfaces.

PROJECTS

TraceForge | Distributed Tracing Pipeline | GitHub

Jul 2026 - Present

Go | Kafka | OpenTelemetry | Docker | Kubernetes | Terraform | In Development

- Building a distributed tracing pipeline in Go that propagates trace context, ingests OpenTelemetry spans, and reliably streams telemetry events through Kafka for asynchronous processing.
- Designing scalable containerized ingestion, processing, and storage services for Kubernetes deployment with Terraform-managed infrastructure and end-to-end observability.
- Defining service-level metrics for ingestion throughput, Kafka consumer lag, and end-to-end trace latency to guide load testing and capacity planning.

Stateful Agentic AI Assistant | GitHub

May 2026 - Jun 2026

LangGraph | Groq | FAISS | Google Embeddings | Docker | AWS EC2 | LangSmith

- Designed a LangGraph-based agent with durable SQLite checkpoints, threaded conversations, streaming, and tool routing for stateful multi-turn interactions across sessions.
- Built a dynamic PDF RAG pipeline with FAISS and Google embeddings; integrated four tools for web search, weather, finance, and calculation with Human-in-the-Loop approval gates.
- Automated Docker build, image publishing, and AWS EC2 deployment through GitHub Actions and a self-hosted runner; traced agent executions with LangSmith for faster debugging.

AI Mock Interview Platform | GitHub

Mar 2026 - Apr 2026

Node.js | Express.js | OpenRouter | Firebase | Razorpay | PDF Reporting

- Built role- and experience-conditioned LLM pipelines that generated adaptive technical and behavioral questions, then produced question-level scores, feedback, and improvement plans.
- Designed six backend REST endpoints covering resume ingestion, question generation, answer submission, session completion, retrieval, and PDF reports; secured Google Sign-In with Firebase.
- Implemented secure Razorpay-based credit purchases, 150-credit onboarding grants, per-session deductions, and downloadable PDF reports to complete the monetization workflow.

Real-Time Collaborative Whiteboard | GitHub

Mar 2025 - May 2025

Node.js | Express.js | Socket.IO | MongoDB | JWT

- Built a room-based collaborative whiteboard that synchronized bidirectional canvas events across clients with sub-second latency and persisted board state in MongoDB.
- Implemented JWT authentication and 10+ APIs for users, sessions, rooms, and board persistence.
- Optimized Socket.IO event handling and MongoDB queries to support 100+ concurrent users, reducing synchronization latency by ~40% and backend response time by ~30%.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL

Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Socket.IO

AI / LLM: LangGraph, RAG, FAISS, Google Embeddings, LangSmith, Groq

Data & Cloud: PostgreSQL, MongoDB, Redis, MySQL, SQLite | AWS, Docker, GitHub Actions, Nginx

Frontend & Tools: React.js, Next.js, Tailwind CSS, Git, GitHub, Firebase, Razorpay

ACHIEVEMENTS

- Earned LeetCode Knight status with a peak rating of 1923 (Top 5.6%) after solving 1,500+ problems and maintaining a 550+ day Problem-of-the-Day streak through consistent daily practice.
- Reached Codeforces Specialist (peak rating 1415) and secured Global Rank 818 in Round 952.
- Selected for Amazon ML Summer School 2024 and placed Top 3 in GCOS 2024 at IIT Kharagpur.
- Secured a Top 5 finish in Overnite at Kshitij, IIT Kharagpur, in a timed algorithmic contest.
- Won IIT Kharagpur's Shyamal Ghosh & Sunanda Ghosh Endowment Honour for 2021-22 academic excellence.